

Surface-independent stability of equilateral and squeezed bispectra on the frozen BIH-VI benchmark: a bounded- η_H all-operator production scan

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July 6, 2026 — draft assembled from locked production data

Abstract

We report a production-scale stability scan of the primordial curvature bispectrum on the frozen BIH-VI benchmark background, an effective-field-theory (EFT) clock realization of an entropy-selected density-shield scenario. The scan applies a validated bounded- η_H all-operator instrument — a finite-basis cubic-action ledger with sealed kernels, independent unit-test engines, and declared hard gates — to equilateral, squeezed ($q = 10^{-3}$), and folded triangle configurations over a hard grid $K \in [0.010, 0.060]$ and a provisional grid $K \in \{0.072, 0.090, 0.119\}$. All 13 declared production gates pass. In the claim-bearing hard envelope $K \leq 0.06$, equilateral and squeezed configurations are stable under the sealed surface-independence (G-S) criteria, with f_{NL} in $[0.02230, 0.03822]$ (equilateral) and $[0.05507, 0.05906]$ (squeezed) at the reference surface. Folded/flattened configurations are diagnostic/report-only under the current sealed instrument; in this run, no folded/flattened row passed the G-FLAT diagnostic. Every reported number is bound to a cryptographically hashed artifact chain with two-party verification; the complete locked data package accompanies this paper. This is an EFT-level methodology and stability result; no UV-completion claim is made.

1 Introduction

Stability of a computed observable under changes of bookkeeping choices — evaluation surface, regulator resolution, operator-basis completion — is a precondition for treating that observable as physical rather than artifactual. This paper reports such a stability program, executed to completion, for the tree-level curvature bispectrum on a single frozen benchmark background of the BIH-VI construction. The calculation is deliberately framed within the standard language of single-clock inflationary perturbation theory and the effective field theory of inflation [1, 2], where the curvature perturbation is the observable clock fluctuation and cubic interactions must be combined with the appropriate constraint and field-redefinition terms. Sharp time-dependent features are known to create localized oscillatory signatures in correlation functions and to stress numerical in-in implementations [3, 4]. The present work uses that setting as a methodology test: a sharp exit transition is allowed, but every reported bispectrum number must survive operator-completion, normalization, convergence, and surface-independence gates before it is treated as claim-bearing.

The background motivation for BIH-VI is an entropy-selected apparent-horizon density shield, in the spirit of horizon-thermodynamic approaches to gravitational dynamics [6, 7]. For observational orientation, the frozen benchmark was chosen from a Planck-compatible scalar/tensor parameter

island [8]. These motivations do not enter as additional claims in this paper: the scan below tests only the stated EFT-clock benchmark and its bounded- η_H bispectrum instrument.

The working discipline throughout is that frame- or scheme-dependent descriptions must be reduced to invariant, operationally measurable statements before they count as physics. Concretely: every claim in this paper is (i) evaluated at multiple hypersurfaces and gated on surface independence, (ii) computed from a sealed operator ledger whose kernels are verified against an independent engine, and (iii) bound to hash-locked artifacts so that each number is reproducible from the archived instrument.

Three boundaries are declared at the outset. First, the BIH-VI benchmark is an EFT-level realization and phenomenological test set, not a final ultraviolet-complete microscopic theory; no UV-completion claim is made. Second, the claim-bearing results are confined to the hard envelope $K \leq 0.06$ in the equilateral and squeezed families; no all-shape closure is claimed. $K \leq 0.06$ is the current claim-bearing boundary of the sealed A2 instrument for the frozen benchmark, not a physical cliff in the EFT-clock model. Third, folded/flattened configurations are reported under a diagnostic-only disposition (Decision A, Section 5); no physical conclusion — neither stability nor instability — is drawn from them.

Nomenclature and definitions

hard envelope The claim-bearing set: equilateral and squeezed hard-grid claim rows with $K \leq 0.06$.

K (grid label) The production-grid label used throughout this paper (the per-mode scale; for equilateral rows $k_1 = k_2 = k_3 = K$). The triangle perimeter $K_\Sigma = k_1 + k_2 + k_3$ is not used as a variable in this paper; where a perimeter is needed it is written explicitly.

sealed G-S criteria The pre-registered G-S surface-independence, frozen-window, Wronskian, convergence, and continuity-sentinel checks, fixed before the production run.

claim-bearing rows Rows included in `claim_rows.csv` and authorized by `A2_ALLOWED_CLAIMS.md`.

non-claim-bearing rows Rows reported for diagnostics, provisional context, or flagged status but excluded from the A2 claim.

folded/flattened rows Diagnostic/report-only unless G-FLAT explicitly passes; in this run, no folded/flattened row passed G-FLAT.

2 Benchmark background

The frozen BIH-VI benchmark is specified by

$$H^2 = H_c^2 y(2 - y), \quad y_N = -m(y)y, \quad m(y) = \mu(1 - y)^p + m_{\text{post}} S(y), \quad (1)$$

with switch profile

$$S(y) = \frac{1}{1 + \exp[(y - y_g)/\Delta]}, \quad (2)$$

and parameters

$$p = 3, \quad \mu = 1, \quad c_s = 1, \quad N_* = 55, \quad y_g = \frac{2}{3}, \quad \Delta = 0.01, \quad m_{\text{post}} = 4, \quad H_c = 3.8 \times 10^{-6} M_{\text{Pl}}. \quad (3)$$

Numerics use the dimensionless convention $H(y_g) = 1$; the value of H_c/M_{Pl} enters only as provenance for physical-amplitude conversions. ($c_s = 1$ is implicit in the mode equation of the sealed instrument; c_s and N_* are fixed by the frozen benchmark specification, hash-referenced in Appendix A.) The background is integrated at $\Delta N = 10^{-3}$ over $N \in [-22, 0.45]$ with fully analytic η and η_N (no numerical differentiation through the switch), and the slow-roll functions (ϵ, η) follow from $m(y)$ as in the sealed instrument.

3 Instrument: the bounded- η_H all-operator ledger

The bispectrum is assembled from the sealed finite-basis ledger

$$B_{\text{total}} = O_1 + O_2 + O_{3a} + O_{3b} + O_{3c} + V_1 + V_2 + f_2 + f_3 + f_4, \quad (4)$$

where $O_1 = a^3 \epsilon^2 \zeta \dot{\zeta}^2$, $O_2 = a \epsilon^2 \zeta (\partial \zeta)^2$, $O_{3a} = -2a \epsilon \dot{\zeta} \partial \zeta \cdot \partial \psi$, $O_{3b} = (\epsilon/2a) \partial \zeta \cdot \partial \psi \partial^2 \psi$, $O_{3c} = (\epsilon/4a) \partial^2 \zeta (\partial \psi)^2$, $V_{1,2}$ are the bounded- η replacement kinetic and gradient terms, and $f_{2,3,4}$ are the nonlocal/decaying field-redefinition contributions. The amplitude is normalized as

$$f_{\text{NL}} = \frac{5}{6} \frac{B_{\text{total}}}{P_1 P_2 + P_1 P_3 + P_2 P_3}, \quad (5)$$

with dimensionful power spectra $P_i = |\zeta_{k_i}|^2$; dimensionless Δ^2 combinations are excluded from all denominators by construction. The following ledger lines are forbidden and absent: O_5 , standalone boundary terms (B_{bdry}) [5], standalone equation-of-motion terms (B_{EOM}), the local ζ^2 field-redefinition line, and any kernel missing its assignment sum. Raw O_4 never enters any physical row or production observable: it is entirely absent from the sealed production script, and in the certified lineage instrument it appears only as an internal diagnostic (the synthetic-context kernel unit test and the regulated IBP-identity gate), as permitted.

Mode functions are integrated with an RK4 scheme from deep inside the horizon ($k/aH \geq 60$) with Bunch–Davies initialization, and bulk integrals carry the sealed $i\epsilon$ regulator ($\epsilon_{\text{reg}} = 0.03$). Kernel implementations are verified against an independent vector/bijection engine (gates UT-K, UT-FR, tolerance 10^{-10}), and the Wronskian identity is enforced along every mode (gate G-W).

4 Validation lineage

The instrument used here is sealed: it is the same code path certified by two prior gate programs, executed and archived under a hash-locked reproduction protocol (Appendix A).

- **G-S surface-independence sweep (v3).** The full ledger was evaluated at five hypersurfaces $N \in \{0.20, 0.25, 0.30, 0.35, 0.40\}$ for squeezed and equilateral rows. Per-row spread gates, a vacuum-pass guard, smallest- K asymptotic requirements, and a drift-scaling slope test were all declared in advance; the certified run passed with six primary rows and established the empirical frozen envelope $K \leq 0.06$ used here.
- **BENCH02 flat-gauge cross-check.** An independent ADHL flat-gauge implementation passed 11 of 11 declared gates against the same benchmark, closing the residual v_3 difference and certifying the template, background, phase, envelope, squeezed, and convergence behavior of the instrument.
- **Production continuity.** The production run verifies continuity sentinels against the certified G-S reference values at $K \in \{0.020, 0.040, 0.060\}$ for both claim families, plus a $\Delta N/2$ convergence sentinel, inside the same gate set (Table 1).

5 Production scan design

The production configuration scans three triangle families — equilateral ($k_1 = k_2 = k_3 = K$), squeezed ($q = k_L/k_S = 10^{-3}$), and folded (near-degenerate, $\delta = 10^{-2}$ with a $\delta = 10^{-3}$ tightening probe) — over the hard grid $K \in \{0.010, 0.015, \dots, 0.060\}$ and the provisional grid $K \in \{0.072, 0.090, 0.119\}$. Thirteen gates are declared: the instrument gates (UT-K, UT-FR, BG-ETA, G-W, CONV), the G-S continuity sentinels, per-family hard-row stability gates, the folded disposition gates (folded rows either pass G-FLAT explicitly or are flagged; no folded claim without a G-FLAT pass), claim-hygiene gates (no provisional claim; no $K > 0.06$ claim), and the execution lock. Per-family row accounting is summarized in Table 3.

Decision A (folded/flattened disposition). Folded/flattened configurations are diagnostic/report-only under the current sealed instrument unless G-FLAT explicitly passes. The G-FLAT diagnostic probes each folded row three ways: regulator halving ($\epsilon_{\text{reg}} \rightarrow \epsilon_{\text{reg}}/2$), flattening-parameter tightening ($\delta = 10^{-2} \rightarrow 10^{-3}$), sign-flip and cancellation-factor screens, in addition to the standard surface-stability and freeze requirements. This disposition was adopted after an earlier wide-grid exploratory scan showed that, near the flattened boundary at the edge of an extended K grid, the ledger sum passes through a zero: the total becomes a small residual of large cancelling operator contributions, and log-slope estimators straddling the sign change report spuriously large scaling exponents. That behavior is an instrument-scope limitation of the sealed kernels in the flattened limit — diagnosed, bounded, and archived — not a physical instability claim; resolving it is deferred to a separately validated instrument revision.

The production scan executed exactly once, under an execution lock requiring explicit authorization flags, from a hash-verified script, with the frozen-artifact record confirmed before and after execution.

6 Results

6.1 Production Run State

The validated bounded- η_H all-operator A2 instrument (script SHA-256 35fd365916395f17...) was applied once to the frozen BIH-VI benchmark under Decision A. The production run (A2_PRODUCTION_LOCAL) completed with verdict PASS in 7.280172109603882 s, scanning 42 shape rows over the hard grid $K \in [0.01, 0.06]$ (step 0.005) and the provisional grid $K \in \{0.072, 0.090, 0.119\}$ for the equilateral, squeezed ($q = 10^{-3}$), and folded families.

6.2 Gate Summary

All 13 declared gates passed (Table 1): kernel unit tests (UT-K, UT-FR $\leq 10^{-10}$), background consistency (BG-ETA), the Wronskian identity (G-W), $\Delta N/2$ convergence (CONV), continuity sentinels against the certified G-S reference values, per-family hard-row stability, and the four claim-hygiene gates (no folded claim without G-FLAT, no provisional claim, no $K > 0.06$ claim, execution lock).

6.3 Claim-Bearing Rows

The claim-bearing set (Table 2) comprises 22 rows: 11 equilateral and 11 squeezed hard-envelope configurations with $K \leq 0.06$, each frozen at all five evaluation surfaces and stable under the sealed G-S criteria. Over this envelope f_{NL} spans $[0.02230, 0.03822]$ (equilateral, mean 0.03146) and $[0.05507, 0.05906]$ (squeezed, mean 0.05769).

6.4 Folded/Flattened Diagnostics

Folded/flattened configurations are diagnostic/report-only under the current sealed instrument (Decision A). Of 14 G-FLAT diagnostic evaluations, 0 passed; all folded rows are therefore reported with full diagnostics (Table 4) and excluded from claims. No physical conclusion is drawn from flagged folded rows.

6.5 Scope and Forbidden Claims

The validated bounded- η_H all-operator A2 instrument was applied to the frozen BIH-VI benchmark. In the claim-bearing hard envelope $K \leq 0.06$, equilateral and squeezed configurations are stable under the sealed G-S criteria. Folded/flattened configurations are diagnostic/report-only under the current sealed instrument; in this run, no folded/flattened row passed G-FLAT.

This is an EFT-level A2 production scan; no UV-completion claim is made. The following claims are explicitly not made: full all-shape closure; folded/flattened physical instability; BIH-VI observational detection; Hubble-tension solution.

7 Scope, limitations, and outlook

The allowed claim of this paper, in full, is: *The validated bounded- η_H all-operator A2 instrument was applied to the frozen BIH-VI benchmark. In the claim-bearing hard envelope $K \leq 0.06$, equilateral and squeezed configurations are stable under the sealed G-S criteria. Folded/flattened configurations are diagnostic/report-only under the current sealed instrument; in this run, no folded/flattened row passed G-FLAT.*

The following claims are explicitly *not* made: full all-shape closure; folded/flattened physical instability; BIH-VI observational detection; UV completion; Hubble-tension solution. Nothing in this paper uses parked interpretive material as evidence. The results are confined to one frozen benchmark background and to the stated envelope; extension of the claim envelope, or a claim-bearing treatment of the flattened limit, requires a separately validated instrument revision and its own gate program. No interpretive C4/C5/C6 material is used in this manuscript. Such material remains outside the load-bearing A2 analysis and is reserved for separate future work.

This audit is specific to the frozen BIH-VI EFT-clock benchmark, the bounded- η_H all-operator ledger, and the declared A2 regulator/validation contract. Comparisons with other sharp-feature calculations require matching conventions, basis choices, field redefinitions, power-spectrum normalization, and regulator prescriptions before numerical discrepancies are interpreted physically.

A Reproducibility and artifact chain

Every number in this paper derives from a hash-locked artifact chain, archived in an append-only ledgered hub and verified by two independent parties (generation-side audit and an external reviewer recomputation, agreeing on all hashes). The chain is:

artifact	role	SHA-256 (truncated)
benchmark handoff specification	frozen benchmark definition	ceba4ebc...7288fad2
a2_gs_surface_sweep_v3.py	certified G-S instrument	3289ccb4...4e48daed
bench02_adhl_flat.py	certified flat-gauge cross-check	b1071166...ac4fa4646
a2_production_scan.py	sealed production script	35fd3659...58c78aae8
production output package (zip)	primary run record	e8500cc3...d2251c266
final data lock package (zip)	frozen, reviewed data release	061dc918...8372cfbe1a

Full values:

```
spec: ceba4ebce5a5ce9d7fc7eaf44ff35f304f1c91cb1e079eb67df532347288fad2
gs_v3: 3289ccb4d7c0443e335c4b7d01d40946770a484806eb713cf33bed394e48daed
bench02: b10711663374cd7e269bec7f83b3b1847e353eb925bd67f46d4f4f5ac4fa4646
production_scan: 35fd365916395f171fee9278480bd102b267f545fb389d6039cfe0058c78aae8
production_zip: e8500cc3db9152200a9ab6242bb22c35c26589976425ccfa978e171d2251c266
data_lock_zip: 061dc91860f355355fc503d36571e4bf571d70dcacdf1aa0c531fa8372cfbe1a
```

Full 64-hexit hashes for every file *inside* the data release, including all tables printed here, are recorded in `SHA256SUMS.txt` and `A2_ARTIFACT_MANIFEST.csv` within the final data lock package, which constitutes the data-availability release for this paper. The lineage-artifact and package-level hashes above are recorded in the hub’s append-only audit ledgers and in the provenance record accompanying this manuscript; the production run wrote its own ledger line at completion.

B Tables

Table 1: Production gate summary (13 gates).

gate	status	key value
UT_K	PASS	value=2.66e-15
UT_FR	PASS	value=1.09e-15
BG_ETA	PASS	etaN_grad_rel=8.7e-05
G_W	PASS	worst_surface_region=2.15e-06
CONV	PASS	rel_dev=0.000841
G_S_CONTINUITY_SENTINELS	PASS	
EQUILATERAL_HARD_ROWS_STABLE	PASS	n=11
SQUEEZED_HARD_ROWS_STABLE	PASS	n=11
FOLDED_GFLAT_OR_FLAGGED	PASS	n=14
NO_FOLDED_CLAIM_WITHOUT_GFLAT	PASS	
NO_PROVISIONAL_CLAIM	PASS	
NO_K_GT_0P06_CLAIM	PASS	
EXECUTION_LOCK	PASS	

Table 2: Claim-bearing rows: equilateral and squeezed hard envelope, $K \leq 0.06$.

family	shape_id	K	$f_{\text{NL}}@0.35$	f_{NL} spread	B drift	status
equilateral	equilateral_K0.010	0.010	0.0382177	2.272e-05	5.744e-04	claim
equilateral	equilateral_K0.015	0.015	0.0361488	5.104e-05	1.367e-03	claim
equilateral	equilateral_K0.020	0.020	0.0370021	9.068e-05	2.369e-03	claim
equilateral	equilateral_K0.025	0.025	0.034999	1.415e-04	3.914e-03	claim
equilateral	equilateral_K0.030	0.030	0.0344472	2.035e-04	5.720e-03	claim
equilateral	equilateral_K0.035	0.035	0.0321143	2.767e-04	8.350e-03	claim
equilateral	equilateral_K0.040	0.040	0.0309027	3.609e-04	1.132e-02	claim
equilateral	equilateral_K0.045	0.045	0.0290793	4.562e-04	1.522e-02	claim
equilateral	equilateral_K0.050	0.050	0.0259711	5.624e-04	2.102e-02	claim
equilateral	equilateral_K0.055	0.055	0.0249104	6.798e-04	2.647e-02	claim
equilateral	equilateral_K0.060	0.060	0.0223037	8.079e-04	3.512e-02	claim
squeezed	squeezed_q0.001_K0.010	0.010	0.0573306	1.857e-05	3.016e-04	claim
squeezed	squeezed_q0.001_K0.015	0.015	0.0583783	4.173e-05	6.644e-04	claim
squeezed	squeezed_q0.001_K0.020	0.020	0.0588645	7.408e-05	1.169e-03	claim
squeezed	squeezed_q0.001_K0.025	0.025	0.0590595	1.156e-04	1.816e-03	claim
squeezed	squeezed_q0.001_K0.030	0.030	0.0588911	1.662e-04	2.619e-03	claim
squeezed	squeezed_q0.001_K0.035	0.035	0.0585432	2.259e-04	3.581e-03	claim
squeezed	squeezed_q0.001_K0.040	0.040	0.0581061	2.946e-04	4.706e-03	claim
squeezed	squeezed_q0.001_K0.045	0.045	0.057511	3.723e-04	6.011e-03	claim
squeezed	squeezed_q0.001_K0.050	0.050	0.0568472	4.590e-04	7.499e-03	claim
squeezed	squeezed_q0.001_K0.055	0.055	0.0559888	5.546e-04	9.204e-03	claim
squeezed	squeezed_q0.001_K0.060	0.060	0.0550745	6.591e-04	1.113e-02	claim

Table 3: Per-family production summary.

family	rows_tested	hard_rows	provisional_rows	claim_rows	flagged_rows	worst_fNL_spread	worst_B_spread_rel	max_abs_fNL_ref035
equilateral	14	11	3	11	3	0.003127	0.2604	0.03822
squeezed	14	11	3	11	3	0.00255	0.05894	0.05906
folded	14	11	3	0	14	0.002033	0.02211	0.09448

Table 4: Folded/flattened G-FLAT diagnostics (report-only under Decision A).

shape_id	K	fNL_ref035	fNL_ieps_half	fNL_delta_tight	instrument_dev	delta_dev	cancellation_factor	sign_flip	gflat_pass	gflat_reason
folded_delta0.01_K0.010	0.01	0.08988	0.275	0.0759	0.6732	0.1555	1.602	False	False	flat_limit
folded_delta0.01_K0.015	0.015	0.09192	0.2818	0.07751	0.6738	0.1567	1.669	False	False	flat_limit
folded_delta0.01_K0.020	0.02	0.09144	0.2756	0.0788	0.6682	0.1382	1.777	False	False	flat_limit
folded_delta0.01_K0.025	0.025	0.09324	0.2849	0.07723	0.6727	0.1718	1.829	False	False	flat_limit
folded_delta0.01_K0.030	0.03	0.0929	0.2822	0.07993	0.6708	0.1396	1.926	False	False	flat_limit
folded_delta0.01_K0.035	0.035	0.09404	0.2894	0.08158	0.6751	0.1325	1.988	False	False	flat_limit
folded_delta0.01_K0.040	0.04	0.09358	0.2876	0.08049	0.6746	0.1399	2.086	False	False	flat_limit
folded_delta0.01_K0.045	0.045	0.09347	0.2884	0.08017	0.6759	0.1423	2.177	False	False	flat_limit
folded_delta0.01_K0.050	0.05	0.09448	0.2965	0.07786	0.6814	0.1759	2.238	False	False	flat_limit
folded_delta0.01_K0.055	0.055	0.09301	0.29	0.07943	0.6792	0.146	2.367	False	False	flat_limit
folded_delta0.01_K0.060	0.06	0.09301	0.2927	0.07958	0.6823	0.1443	2.457	False	False	flat_limit
folded_delta0.01_K0.072	0.072	0.09321	0.302	0.07646	0.6914	0.1797	2.672	False	False	flat_limit
folded_delta0.01_K0.090	0.09	0.08963	0.2951	0.07585	0.6962	0.1537	3.144	False	False	flat_limit
folded_delta0.01_K0.119	0.119	0.0837	0.2882	0.06904	0.7096	0.1752	4.032	False	False	flat_limit

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